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Stress Analysis And Design Optimization Of A Pressure Vessel Using Ansys Package

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Abstract

The pressure vessels are closed containers used to hold gases or liquids at a pressure substantially different from the ambient pressure. The pressure inside the vessel is different and may changes by the conditions. The vessels are too dangerous and fatal accidents have occurred in the history of pressure vessel development and operation. Accordingly vessel design, manufacture, and operation are regulated by engineering authorities backed by legislation.

The main objective of this paper is to design and analysis of pressure vessel. While designing various parameters of Pressure Vessel checked and designed according to the principles specified in American Society of Mechanical Engineers [1] (A.S.M.E) Sec VIII Division 1.

The stress development in the pressure wall critical points are analyzed by using Ansys 15 and an optimized model is modelled to overcome the stresses produced in the vessel. For designing we used Pro-E Creo. Both model parts are [2] designed and analyzed individually with ASME three different materials with the results the best suitable material with good design is developed.

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Keywords: ASME, Ansys, Creo, head, Pressure vessel and shell ;

1. Introduction

Pressure vessels are used to store fluids under pressure. The fluid being stored may undergo a change of state inside the pressure vessel as in case of steam boilers or it may combine with other reagents as in a chemical plant. The pressure vessels are designed with great care because rupture of pressure vessels means an explosion which may cause loss of life and property. The material of pressure vessels may be brittle such that cast iron or ductile such as mild steel. Cylindrical or spherical pressure vessels are commonly used in industry to carry both liquids and gases under pressure. When the pressure vessel is exposed to this pressure, the material comprising the vessel is subjected to pressure loading, and hence stresses, from all directions. The normal stresses resulting

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from this pressure are functions of the radius of the element under consideration, the shape of the pressure vessel as well as the applied pressure.

1.1 Amine pressure vessel

Amine gas treating also known as amine scrubbing, gas sweetening and acid gas removal, refers to a group of processes that use aqueous solutions of various alkylamines (commonly referred to simply as amines) to remove hydrogen sulfide (H_2S) and carbon dioxide (CO_2) from gases. It is a common unit process used in refineries, and is also used in petrochemical plants, natural gas processing plants and other industries.

Processes within oil refineries or chemical processing plants that remove hydrogen sulfide are referred to as "sweetening" processes because the odor of the processed products is improved by the absence of hydrogen sulfide. An alternative to the use of amines involves membrane technology. However, membrane separation is less attractive due to the relatively high capital and operating costs as well as other technical factors.

Many different amines are used in gas treating:

- Diethanolamine (DEA)
- Monoethanolamine (MEA)
- Methyldiethanolamine (MDEA)
- Diisopropanolamine (DIPA)
- Aminoethoxyethanol (Diglycolamine) (DGA)

The most commonly used amines in industrial plants are the alkanolamines DEA, MEA, and MDEA shown in figure 1.1. These amines are also used in many oil refineries to remove sour gases from liquid hydrocarbons such as liquefied petroleum gas (LPG).

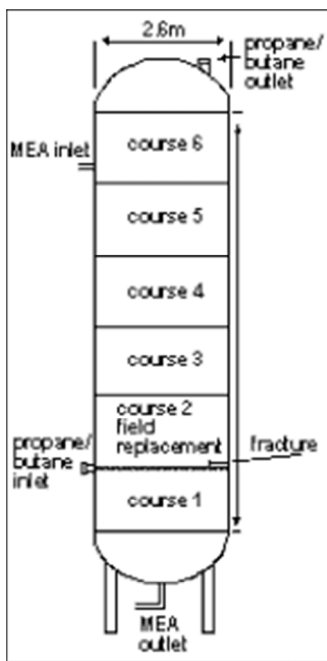


Fig 1.1 Amine pressure vessel

1.2 Component of pressure vessel

There are three main Type are given below

1. Shell
2. Head
3. Nozzle

2. Design criteria [4]:

2.1 Factors influencing the design

Regardless of the nature of application of the vessels, a number of factors usually must be considered in designing the unit. The most important consideration often is the selection of the type of vessel that performs the required services in the most satisfactory manner. In developing the design, a number of other criteria must be considered such as the properties of material used, the induced stresses, the elastic stability, and the aesthetic appearance of the unit. The cost of fabricated vessel is also important in relation to its service and useful life [3].

2.2 Design of pressure vessels to code specification

[10] American, Indian, British, Japanese, German and many other codes are available for design of pressure vessels. However the internationally accepted for design of pressure vessel code is American Society of Mechanical Engineering (ASME). Various codes governing the procedures for the design, fabrication, inspection, testing and operation of pressure vessels have been developed; partly as safety measure. These procedures furnish standards by which, any state can be assured of the safety of pressure vessels installed within its boundaries. The code used for unfired pressure vessels is Section VIII of the ASME boiler and pressure vessel code. It is usually necessary that the pressure vessel equipment be designed to a specific code in order to obtain insurance on the plant in which the vessel is to be used. Regardless of the method of design, pressure vessels within the limits of the ASME code specification are usually checked against these specifications [5].

2.3 Input data [7]

- Material: SA516 Gr60
- Max Allowable Stress, $S = 118 \text{ MPa}$
- Design Pressure, $P = 3.434 \text{ Mpa}$
- Design Temperature, $T = 343 \text{ K}$
- Ellipsoidal top head is selected. [8] (According to UG 31 of ASME Section VIII Div 1)

[11] As per ASME standard calculations the output dimensions of the pressure vessels are shown in following figures 2.3.1 to 2.3.5.

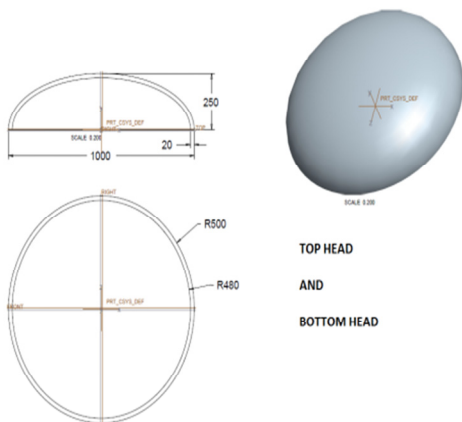


Fig 2.3.1 Top head and bottom head of pressure vessel

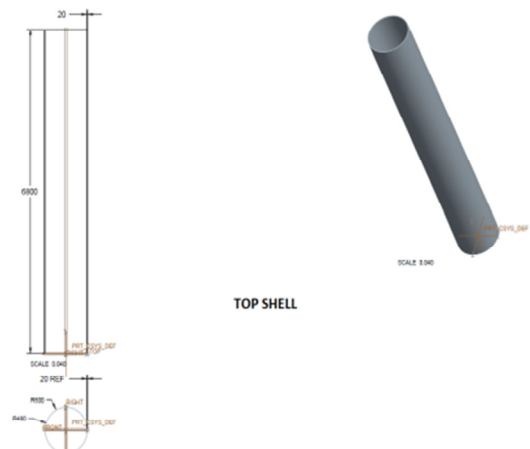
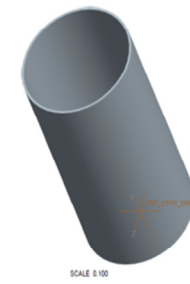
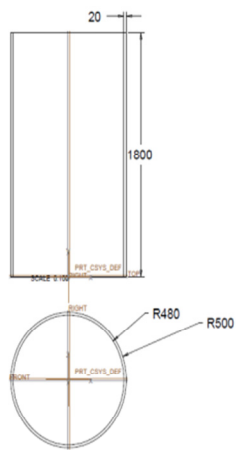
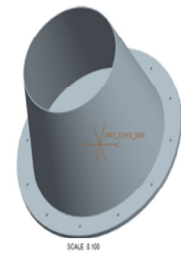
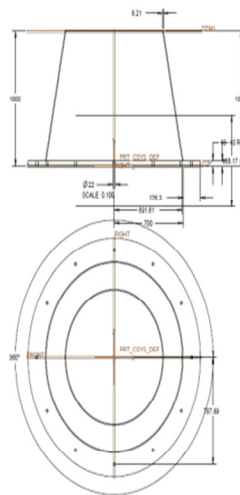


Fig 2.3.2 Top shell of pressure vessel



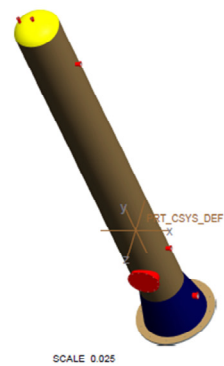
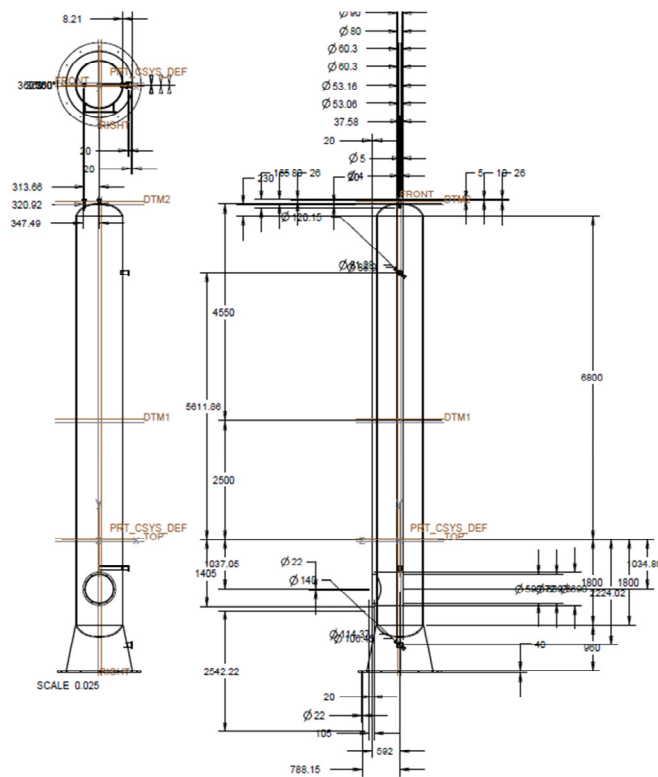
BOTTOM SHELL



SKIRT

Fig 2.3.3 Bottom shell of pressure vessel

Fig 2.3.4 Skirt model of pressure vessel



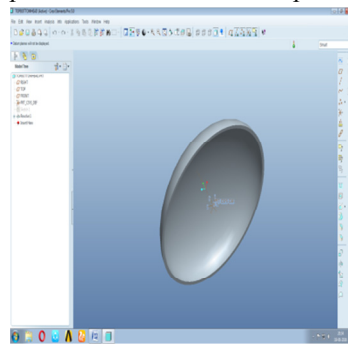
SCALE 0.025

PRESSURE VESSEL

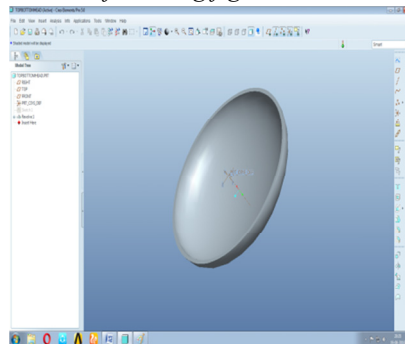
Fig 2.3.5 Pressure vessel model

3. Modelling by using pro-e creo [12]:

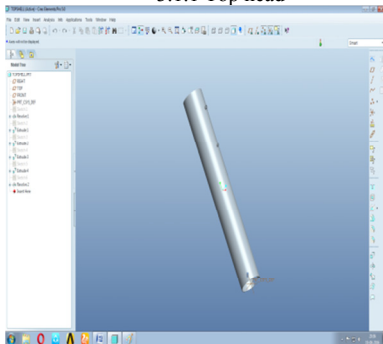
3.1 Actual pressure vessel individual parts modelling are shown in following figures 3.1.1 to 3.1.4.



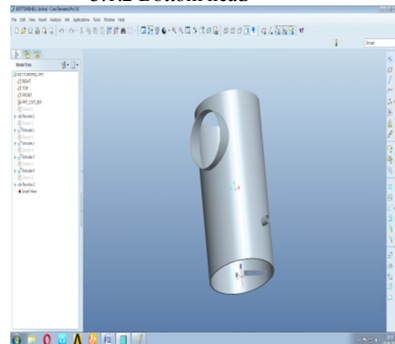
3.1.1 Top head



3.1.2 Bottom head

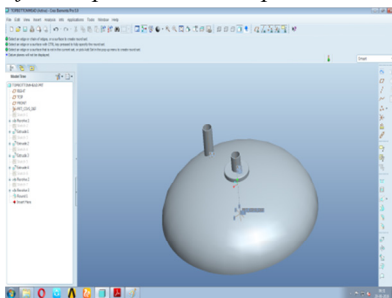


3.1.3 Top shell

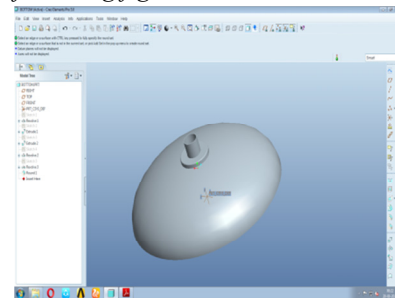


3.1.4 Bottom shell

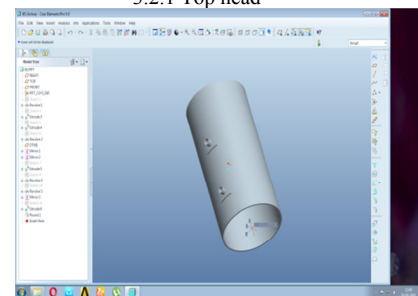
3.2 Reinforced pressure vessel parts modelling shown in following figures 3.2.1 to 3.2.4.



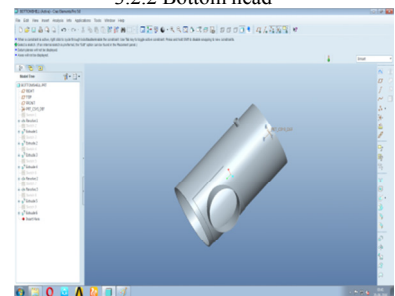
3.2.1 Top head



3.2.2 Bottom head



3.2.3 Top shell



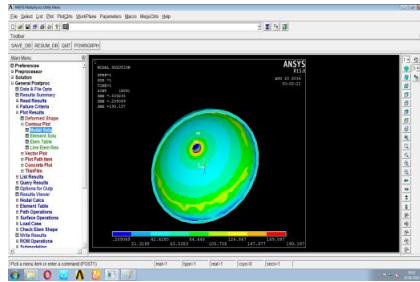
3.2.4 Bottom shell

4. Results and discussion [9]:

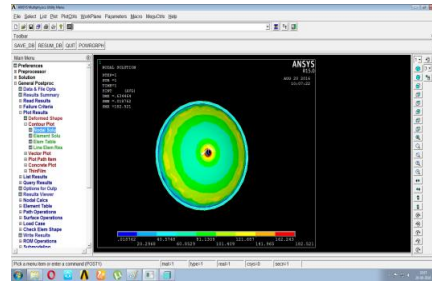
4.1 Model: Bottom head

The [6] stress distribution in the bottom head of the pressure are observed in the following figures 4.1.1 to 4.1.3

4.1.1 Material: SA516GR60

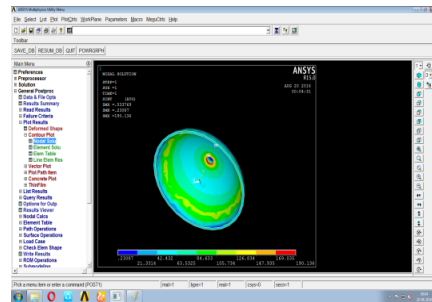


4.1.1(a) Actual model stress intensity

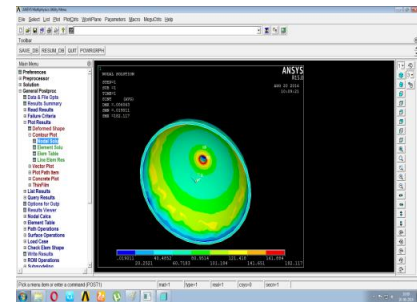


4.1.1(b) Reinforced model stress intensity

4.1.2 Material: SA516GR70

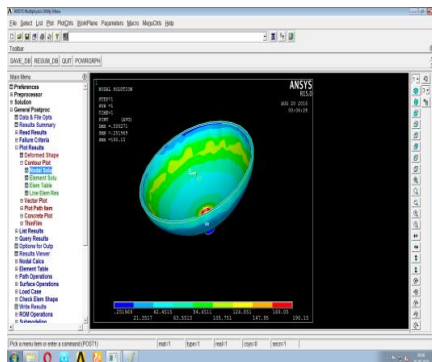


4.1.2(a) Actual model stress intensity

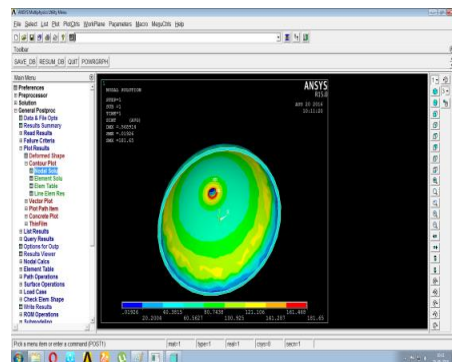


4.1.2(b) Reinforced model stress intensity

4.1.3 Material: SA516GR65



4.1.3(a) Actual model stress intensity

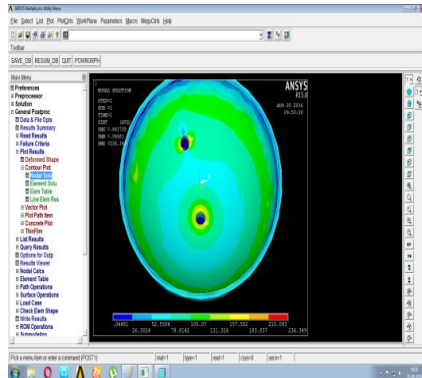


4.1.3(b) Reinforced model stress intensity

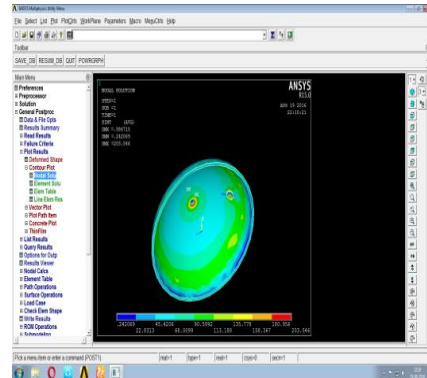
4.2 Model: Top head

The stress distribution in the top head of the pressure are observed in the following figures 4.2.1 to

4.2.1 Material: SA516GR60

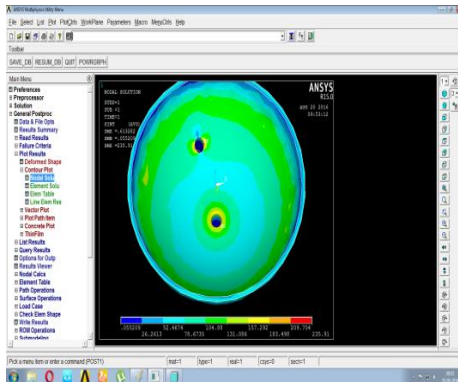


4.2.1(a) Actual model stress intensity

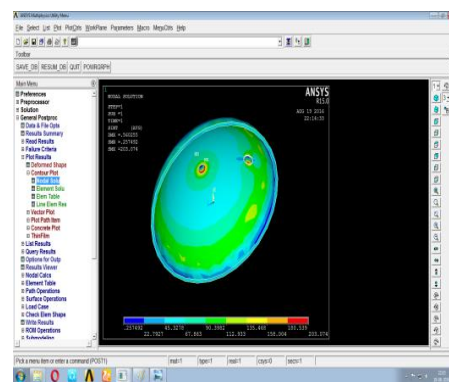


4.2.1(b) Reinforced model stress intensity

4.2.2 Material: SA516GR70

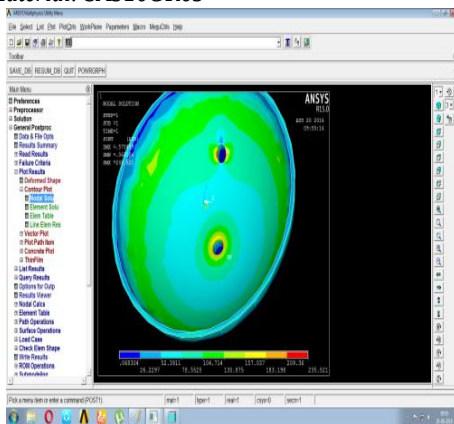


4.2.2(a) Actual model stress intensity

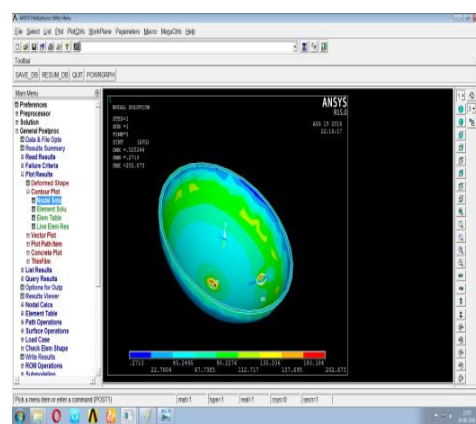


4.2.2(b) Reinforced model stress intensity

4.2.3 Material: SA516GR65



4.2.3(a) Actual model stress intensity

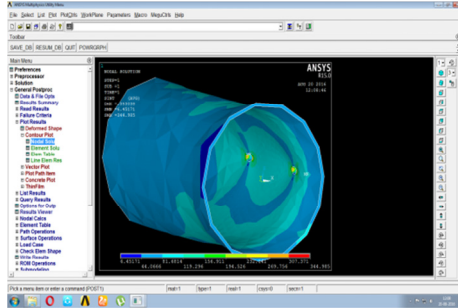


4.2.3(b) Reinforced model stress intensity

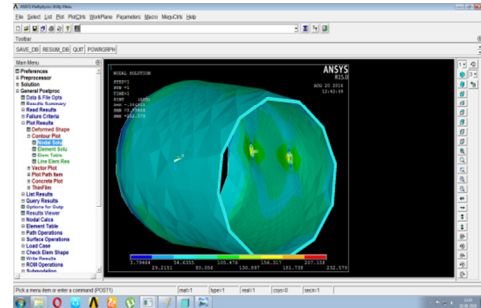
4.3 Model: Top shell

The stress distribution in the top shell of the pressure are observed in the following figures 4.3.1 to 4.3.3

4.3.1 Material: SA516GR60

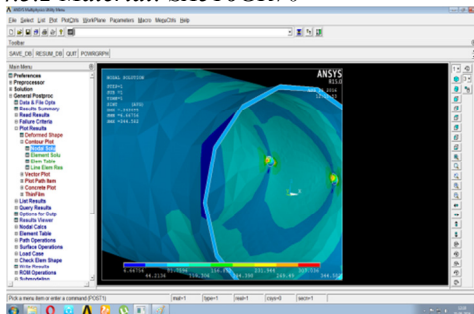


4.3.1(a) Actual model stress intensity

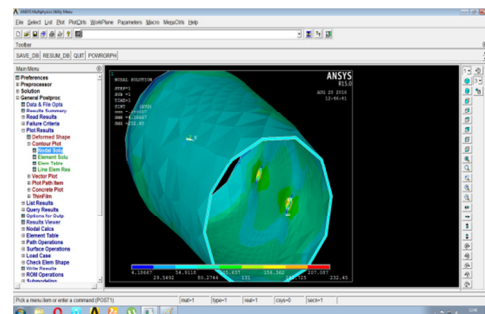


4.3.1(b) Reinforced model stress intensity

4.3.2 Material: SA516GR70

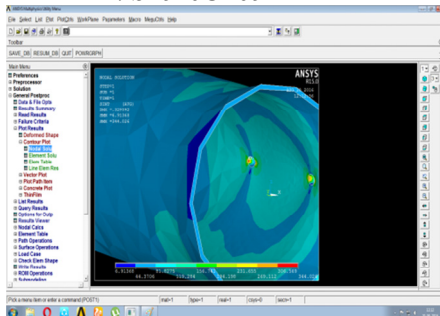


4.3.2(a) Actual model stress intensity

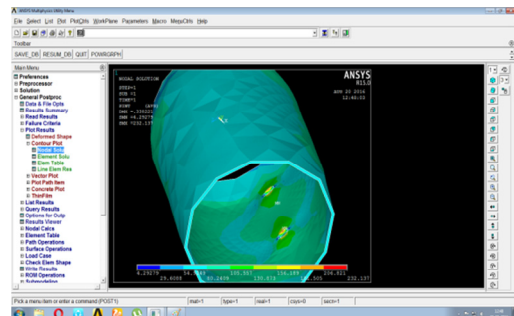


4.3.2(b) Reinforced model stress intensity

4.3.3 MATERIAL: SA516GR65



4.3.3(a) Actual model stress intensity

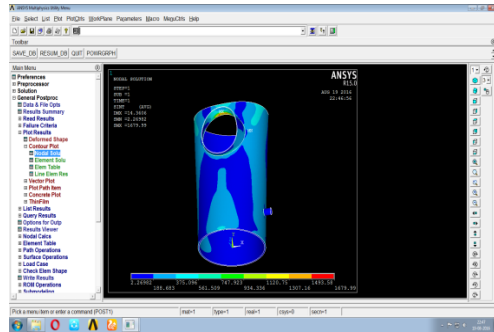


4.3.3(b) Reinforced model stress intensity

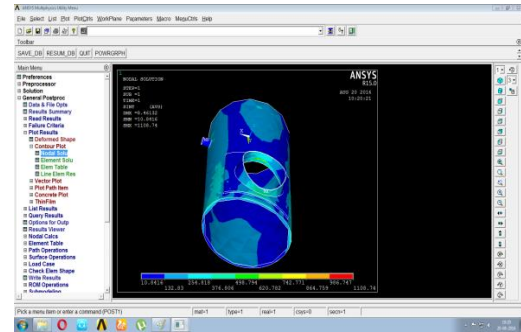
4.4 Model: Bottom shell

The stress distribution in the bottom shell of the pressure are observed in the following figures 4.4.1 to 4.4.3

4.4.1 Material: SA516GR60

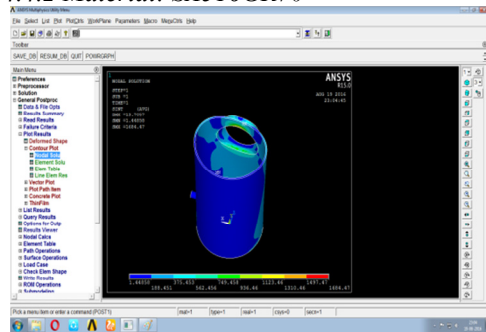


4.4.1(a) Actual model stress intensity

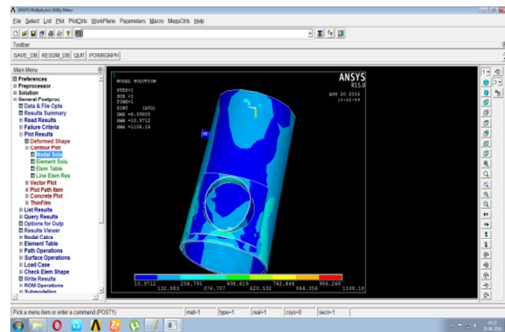


4.4.1(b) Reinforced model stress intensity

4.4.2 Material: SA516GR70

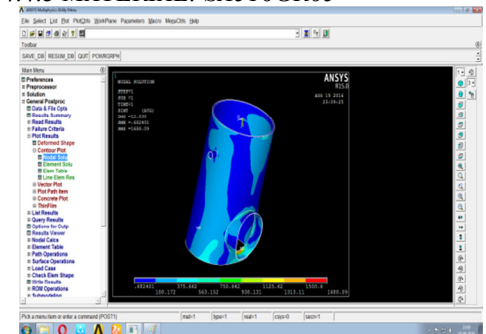


4.4.2(a) Actual model stress intensity

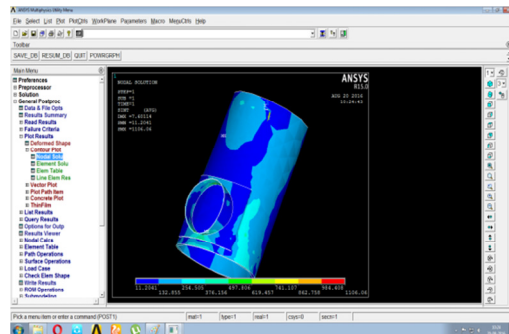


4.4.2(b) Reinforced model stress intensity

4.4.3 MATERIAL: SA516GR65



4.4.3(a) Actual model stress intensity



4.4.3(b) Reinforced model stress intensity

5. Summary of results

The overall stress distribution in all the parts are tabulated in the following tables 5.1 to 5.4

Table 5.1 Top head results

S.NO	Materials	Stress intensity	
		Actual model	Reinforced model
1	SA516GR60	236.349	203.546
2	SA516GR70	235.91	203.074
3	SA516GR65	235.521	202.673

Table 5.2 Bottom head

S.NO	Materials	Stress intensity	
		Actual model	Reinforced model
1	SA516GR60	190.197	182.521
2	SA516GR70	190.136	182.117
3	SA516GR65	190.15	181.65

Table 5.3 Top shell

S.NO	Materials	Stress intensity	
		Actual model	Reinforced model
1	SA516GR60	344.985	232.579
2	SA516GR70	344.582	232.45
3	SA516GR65	344.026	232.137

Table 5.4 Bottom shell

S.NO	Materials	Stress intensity	
		Actual model	Reinforced model
1	SA516GR60	1688.09	1108.74
2	SA516GR70	1684.47	1108.18
3	SA516GR65	1679.99	1106.06

6. Conclusion

Designing the parts of the vessel as per the ASME is successful. The analysis is performed on both model parts with three different materials and the results are compared.

By the results we observed that the stress and deformed values of the modified model is better than the actual model because of the reduction of the stress values the life time of the vessel may increase.

In the analysis part we used three materials as per the regulations of ASME in observation of results SA516GR65 has the stress values less compared to all other materials by this we conclude that the for pressure vessel design SA516GR65 may be used and with the small modifications in design the life time of the pressure vessel may increases.

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